

Vol 10 Architectural Scaffold v0.1 — Mathematical Methods for BST

Filed: 2026-05-23 Saturday (Lyra, Wave 2 scaffold) Companion to: INDEX.md

Anchoring BST results (existing coverage by chapter)

Ch 1 Linear Algebra + Hilbert Spaces (~60%)

Anchors: Vol 1 Ch 2 + Bergman $H^2(D_{IV}^5)$ framework - Standard L^2 + inner product + bounded operators - BST cross-link: Bergman $H^2(D_{IV}^5)$ is the substrate's canonical Hilbert space

Ch 2 Complex Analysis + Holomorphic Functions (~50%)

Anchors: Vol 0 Ch 1 + $D_{IV}^5 \subset \mathbb{C}^5$ - Standard complex analysis + Cauchy's integral formula - BST cross-link: D_{IV}^5 as bounded holomorphic substrate; Bergman reproducing kernel

Ch 3 ODEs + Sturm-Liouville (~50%)

Anchors: Vol 1 Ch 7 dynamics + Vol 5 Ch 4 - Standard ODE methods + boundary-value problems - BST cross-link: substrate Schrödinger as Sturm-Liouville on Bergman H^2

Ch 4 PDEs + Heat / Wave / Laplace (~50%)

Anchors: Vol 4 Ch 5 partition function + heat kernel chain - Standard PDE methods + Green's functions - BST cross-link: heat kernel evaluation on D_{IV}^5 (Vol 6 Ch 5 LOAD-BEARING)

Ch 5 Fourier Analysis + Distributions (~50%)

Anchors: Wallach K-type decomposition + Vol 1 Ch 5 - Standard Fourier transforms + delta-function distributions - BST cross-link: K-type decomposition of Bergman H^2 is substrate-Fourier

Ch 6 Group Theory + Lie Groups (~60%)

Anchors: $SO_0(5,2)$ + Vol 0 Ch 4 + Vol 1 Ch 5 - Standard finite + Lie group theory + Casimir invariants - BST cross-link: $SO_0(5,2)$ substrate group + Casimir algebra (Vol 1 Ch 5)

Ch 7 Differential Geometry (~50%)

Anchors: Vol 0 Ch 1 + D_{IV}^5 bounded Hermitian symmetric domain - Standard manifolds + connections + Riemann curvature - BST cross-link: D_{IV}^5 bounded Hermitian symmetric domain + Bergman metric

Ch 8 Representation Theory (~60%)

Anchors: Wallach 1976 + Vol 1 Ch 2 - Standard finite + compact group representation theory - BST cross-link: Wallach 1976 K-type classification of Bergman $H^2(D_{IV}^5)$

Ch 9 Special Functions (~50%)

Anchors: May Program investigation (Fox H + Aleph + Heckman-Opdam + mock theta) - Standard Bessel + Legendre + hypergeometric functions - BST cross-link: Substrate-natural special functions per May Program; $\zeta_B(s)$ substrate-spectral zeta

Ch 10 Calculus of Variations + Path Integrals (~50%)

Anchors: Vol 1 Ch 7 dynamics + Feynman path integral - Standard variational principles + Euler-Lagrange - BST cross-link: Substrate path integral via substrate-tick $GF(128)^k$ finite-step sum

Ch 11 Asymptotic Analysis + WKB (~40%)

Anchors: Vol 1 Ch 10 substrate-tick UV + $\alpha^{\{\text{BST primary}\}}$ pattern (T2476) - Standard asymptotic methods + WKB approximation - BST cross-link: substrate-tick UV limits + T2476 $\alpha^{\{\text{BST primary}\}}$ exponent pattern

Ch 12 Numerical Methods + Computation (~70%)

Anchors: BST verification toys + Toy 541 + Toy 639 + `verify_bst.py` - Standard numerics + Monte Carlo + spectral methods - BST cross-link: `python3 play/verify_bst.py` reproduction package; ~3500 toys

~55% existing implicit coverage (BST research uses standard methods throughout); Vol 10 surfaces methods explicitly for mathematicians; 45% pedagogical exposition to build.

Cross-volume reference map

Inputs from all volumes: Vol 10 is the reference for mathematical techniques used throughout BST Outputs to: Mathematician readers; cross-reference manual for all BST volumes

— Vol 10 Architectural Scaffold v0.1, Saturday 2026-05-23 Wave 2 scaffold